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Functional polymer brushes *via* surface-initiated atom transfer radical graft polymerization for combating marine biofouling

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Dense and uniform polymer brush coatings were developed to combat marine biofouling. Nonionic hydrophilic, nonionic hydrophobic, cationic, anionic and zwitterionic polymer brush coatings were synthesized *via* surface-initiated atom transfer radical polymerization (SI-ATRP) of 2-hydroxyethyl methacrylate, 2,3,4,5,6-pentafluorostyrene, 2-(methacryloyloxy)ethyl trimethylammonium chloride, 4-styrenesulfonic acid sodium and *N,N'*-dimethyl-(methacryloyl ethyl) ammonium propanesulfonate, respectively. The functionalized surfaces had different efficacies in preventing adsorption of bovine serum albumin (BSA), adhesion of the Gram-negative bacterium *Pseudomonas* sp. NCIMB 2021 and the Gram-positive *Staphylococcus aureus*, and settlement of cyprids of the barnacle *Amphibalanus amphitrite* (= *Balanus amphitrite*). The nonionic hydrophilic, anionic and zwitterionic polymer brushes resisted BSA adsorption during a 2 h exposure period. The nonionic hydrophilic, cationic and zwitterionic brushes exhibited resistance to bacterial fouling (24 h exposure) and cyprid settlement (24 and 48 h incubation). The hydrophobic brushes moderately reduced protein adsorption, and bacteria and cyprid settlement. The anionic brushes were least effective in preventing attachment of bacteria and barnacle cyprids. Thus, the best approach to combat biofouling involves a combination of nonionic hydrophilic and zwitterionic polymer brush coatings on material surfaces.

Keywords: polymer brushes; antifouling; microfouling; macrofouling; SI-ATRP and barnacles

Introduction

Biofouling, defined as the undesirable accumulation of biomolecules, microorganisms, plants and animals on artificial surfaces, has adverse effects on materials immersed in water (Yebra et al. 2004; Chambers et al. 2006; Grozea and Walker 2009). The process starts with adsorption of organic molecules. Then, depending upon conditions in the water, microfouling and/or macrofouling will occur on the surfaces of the substratum (Clare et al. 1992; Rittschof 2000; Almeida et al. 2007). To control and manage biofouling, a variety of antifouling (AF) coatings have been developed. The most common at present are coating systems that use metallic species, such as copper and zinc, in modified self-polishing copolymer paints. However, their toxicity to aqueous organisms and the environment are under scrutiny (Chambers et al. 2006; Almeida et al. 2007). Therefore, development of environmentally-benign technologies, instead of coatings that leach broad spectrum metals or organic biocides, is of great

importance and interest (Rittschof 2000; Magin et al. 2010; Callow and Callow 2011).

Non-toxic functional polymer brushes have been employed as non-releasing coatings to prevent the attachment of proteins, bacteria and marine organisms (Krishnan et al. 2008; Grozea and Walker 2009; Banerjee et al. 2011). In many anti-biofouling surfaces involving polymer brushes, the synthetic polymers were immobilized on the substrata by the 'grafting to' approach, either *via* physisorption or covalent bonding from the solution (Krishnan et al. 2006; Statz et al. 2006; Tan et al. 2010). As an alternative, surface-initiated (SI) polymerization of functional monomers from substrata (the 'grafting from' approach) allows the preparation of dense and stable polymer brushes (Edmondson et al. 2004; Xu et al. 2009a), which are favorable to the inhibition of biofouling.

Surface properties, such as surface wettability and charges, play a significant role in the microfouling and macrofouling process. Previous studies have reported

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the effect of surface hydrophilicity either on micro-organism adhesion (Baier et al. 1968; Baier and Meyer 1992; Ista et al. 2004; Katsikogianni et al. 2008) or on attachment of macrofoulers (Rittschof and Costlow 1989; Roberts et al. 1991; Gerhart et al. 1992; Holm et al. 1997; Petrone et al. 2011). Apart from wettability, surface charges play important roles in adhesion of macrofoulers (Roberts et al. 1991; Holm et al. 1997; Petrone et al. 2011). The influence of surface charges on microfouling and macrofouling has yet to be thoroughly investigated.

Atom transfer radical polymerization is a controlled/living free-radical polymerization technique, allowing polymerization of a wide range of monomers and initiation from various types of substrata to produce well-defined functional polymer brush coatings (Matyjaszewski and Xia 2001; Xu et al. 2009a). In the present work, well-defined polymer brushes with varying degrees of hydrophilicity and charges were synthesized *via* surface-initiated atom transfer radical polymerization (SI-ATRP) from glass substrata. 2-Hydroxyethyl methacrylate (HEMA) and 2,3,4,5,6-pentafluorostyrene (PFS) were utilized to prepare nonionic hydrophilic and nonionic hydrophobic polymer brush coatings, respectively. 2-(Methacryloyloxy)ethyl trimethylammonium chloride (META), 4-styrenesulfonic acid sodium (NaSS) and *N,N'*-dimethyl-(methylmethacryloyl ethyl) ammonium propanesulfonate (DMAPS) were employed to synthesize polymer brush coatings bearing, respectively, cationic, anionic and zwitterionic functional groups. Cationic polymers were synthesized by quaternization of tertiary amine of polymer brushes with various alkyl halides (Huang et al. 2008; Yuan et al. 2009; Engler et al. 2011). Zwitterionic sulfobetaine polymers were synthesized by modification of poly(2-dimethylamino)-ethyl methacrylate homopolymers with 1,3-propane sulfone (Lowe et al. 1999; Ward et al. 2006; Stacha et al. 2011). Direct synthesis of ionic polymer brushes is preferable. In this study, direct polymerization of ionic monomers from glass substrata was performed by SI-ATRP in protic media, using methanol/water and isopropanol/water mixtures to ensure reasonable monomer/polymer solubility and low polydispersity. The AF performance of the polymer functionalized-surfaces was revealed by the extent of adsorption of bovine serum albumin, adhesion of both the Gram-negative bacterium *Pseudomonas* sp. NCIMB 2021 and the Gram-positive *Staphylococcus aureus*, as well as settlement of cyprids of the macrofouling barnacle *Amphibalanus amphitrite* (= *Balanus amphitrite*). The 'grafting from' approach to the preparation of functional polymer brushes can be readily extended to other substrata of interest to the marine and biomedical industries (Neoh and Kang 2011).

Experimental section

Materials

4-(Chloromethyl)phenyl trichlorosilane (CTS, 97%), 2-hydroxyethyl methacrylate (HEMA, 97%), 2-(methacryloyloxy)ethyl trimethylammonium chloride solution (META, 80 wt.% in H₂O), 4-styrenesulfonic acid sodium salt hydrate (NaSS), *N,N'*-dimethyl-(methylmethacryloyl ethyl) ammonium propanesulfonate (DMAPS, 97%), triethylamine (TEA, 98%), copper(I) chloride (CuCl, 99%), copper(II) chloride (CuCl₂, 97%), *N,N,N',N',N''*-pentamethyldiethylenetriamine (PMDETA, reagent grade), 2,2'-bipyridine (BPY, 99%) and fluorescein isothiocyanate-conjugated bovine serum albumin (BSA-FITC) were purchased from Sigma-Aldrich Chemical Co., St Louis, MO, USA. 2,3,4,5,6-Pentafluorostyrene (PFS, 97%) was obtained from Alfa Aesar Co., Ward Hill, MA, USA. The marine strain of the Gram-negative bacterium *Pseudomonas* sp. NCIMB 2021 was obtained from the National Collection of Marine Bacteria, Sussex, UK and the Gram-positive *Staphylococcus aureus* (*S. aureus* ATCC, 25923) from the American Type Culture Collection, Manassas, VA, USA.

Immobilization of the silane coupling agent on the substratum surface

The glass slides (GS) were first activated by immersing in piranha solution (H₂SO₄ (95–97%):H₂O₂ (30%) = 3:1, v:v) for 30 min to generate a hydroxyl-enriched surface (Yang et al. 2011), denoted as the GS-OH surface. The GS-OH substrata were immersed in 20 ml of anhydrous toluene containing TEA (0.56 ml, 4×10^{-3} mol). CTS (0.4 ml, 2×10^{-3} mol) was added dropwise using a syringe. The reaction was carried out at room temperature for 24 h and the substrata were ultrasonically washed with acetone (3 × 20 ml), ethanol (4 × 20 ml) and deionized water, in that order. The CTS-coupled glass substrata (GS-CTS) were dried in a clean oven at 70°C for 2 h.

Polymer brushes via surface-initiated atom transfer radical polymerization (SI-ATRP)

For the preparation of glass substrata with grafted polymer brushes, the monomer and ligand were added to the solvent containing the GS-CTS substrata. The mixture was stirred and degassed with argon for 30 min prior to the addition of catalyst. The graft polymerization was allowed to proceed for a predetermined period of time. After the reaction, the modified glass substrata were washed thoroughly with 3 × 100 ml of solvent to remove the physically adsorbed monomer and polymer. For the preparation

of HEMA polymer-grafted glass (GS-PHEMA) surface, SI-ATRP was carried out at a [HEMA]:[CuCl]:[CuCl₂]:[PMDETA] molar feed ratio of 100:1:0.2:1 in ethanol at room temperature for 6 h. For the preparation of the PFS polymer-grafted glass (GS-PPFS) surface, the molar feed ratio of [PFS]:[CuCl]:[CuCl₂]:[PMDETA] was 100:1:0.2:1, using *N,N*-dimethylformamide (DMF) as the solvent at 110°C for 24 h. For the preparation of the META polymer-grafted glass (GS-PMETA) surface, SI-ATRP was carried out at a [META]:[CuCl]:[CuCl₂]:[BPY] molar feed ratio of 100:1:0.2:2 in a mixture of isopropanol (IPA) and deionized water (IPA:water = 3:2, v:v) at room temperature for 24 h. SI-ATRP was carried out at a [NaSS]:[CuCl]:[CuCl₂]:[BPY] molar feed ratio of 100:1:0.2:2 in methanol and deionized water (methanol:water = 1:1, v:v) at room temperature for 24 h to prepare the poly(NaSS)-grafted glass (GS-PNaSS) surface. For the preparation of the DMAPS polymer-grafted glass (GS-PDMAPS) surface, SI-ATRP of DMAPS was also carried out at a [DMAPS]:[CuCl]:[CuCl₂]:[BPY] molar feed ratio of 100:1:0.2:2 in a mixture of methanol and deionized water (methanol:water = 4:1, v:v) at room temperature for 24 h. The prepared GS-PDMAPS were washed with a copious amount of phosphate buffered saline (PBS) to remove the physically adsorbed monomer and polymer. For the functionalization of glass vials for barnacle cyprid settlement studies, the glass vials were activated and silanized as the glass slides. The respective monomer, copper catalyst, ligand and solvent were introduced directly into the vials to give a final volume of reaction mixture of ~4 ml. The SI-ATRP process was similar to that described above for the corresponding planar substrata.

The free (unbound) homopolymer was prepared in one-pot by adding benzyl chloride as the free initiator during the synthesis of the corresponding polymer functionalized-surface *via* SI-ATRP. The reaction mixture of nonionic PHEMA homopolymer and the mixture of hydrophobic PPFS homopolymers were purified by passing through a neutral alumina column to remove the copper catalyst. The resulting PHEMA solution was dialyzed in ethanol for 3 days to remove the unreacted monomer, followed by precipitation into diethyl ether to obtain the white solid PHEMA homopolymer. The PPFS homopolymer was precipitated in methanol and purified twice by re-dissolution in tetrahydrofuran and re-precipitation in methanol.

Surface and material characterization

The chemical composition of the polymer-functionalized substrata was determined by X-ray photoelectron spectroscopy (XPS) on a Kratos AXIS Ultra^{DLD}

spectrometer with a monochomatized Al K α X-ray source (1486.71 eV photons). The core-level signals were obtained at the photoelectron take-off angle (α , with respect to the sample surface) of 90°. The AXIS charge neutralisation system provides charge compensation on all types of insulating materials by low energy electrons and therefore does not damage the surface of the sample (Metson 1999). Corrections were made to account for differences in kinetic energy of photoelectrons. The binding energies (E_B) of ejected photoelectrons is calculated from the following equation: $E_K = h\nu - E_B - \Phi$, in which the only measured variable is the kinetic energy (E_K) of the ejected electron and $h\nu$ is the energy of the exciting photon source and Φ is the spectrometer work function. The work function Φ is determined by calibration of the instrument using E_B values for electrons from standard orbitals of elements (C 1s and Au 4f, etc.) (Moulder et al. 1992; Metson 1999). Other corrections to the above equation, such as the atomic recoil energy, are small and can generally be ignored (Metson 1999). All binding energies (BEs) were referenced to the neutral C 1s hydrocarbon peak at 284.6 eV. In peak analysis, a linear baseline correction was applied. The line width (full width at half-maximum, or FWHM) for the Gaussian peaks was maintained constant for all peak components in a particular spectrum.

The topography of polymer-modified substratum and the thickness of the polymer brush coatings were studied by atomic force microscope (AFM) in tapping mode, using a Nanoscope IIIa AFM with a set-point ratio of 0.8. The AFM phase images were also measured to reveal variations in their surface properties, such as composition, adhesion, friction and viscoelasticity. To measure the thickness of the polymer brush coatings, the polymer-functionalized substrata were immersed in liquid nitrogen for 30 s and then scratched using a razor blade. As the glass substratum was much harder than the polymer layer, only the polymer coatings was scratched away and the thickness of the polymer layer was imaged (de Vos et al. 2010). Static water, glycerol and the diiodomethane contact angles (CA) of the uncoated and functionalized surfaces were measured at 25°C, using the sessile drop method with a 2 μ l liquid droplet, in a telescopic goniometer (model 100-00-(230), Rame-Hart, Inc., Mountain Lake, NJ, USA). The surface energy (γ_s) for each surface was calculated using Antonow's empirical equation (Antonow 1907; Kwok and Neumann 1999): $\gamma_s = \gamma_L(1 + \cos \theta)/2$, where γ_L is the interfacial tension between the liquid and vapour phases ($\gamma_L = 72.8 \text{ mJ m}^{-2}$ for water and air, $\gamma_L = 64 \text{ mJ m}^{-2}$ for glycerol and air, and $\gamma_L = 64 \text{ mJ m}^{-2}$ for diiodomethane and air) (Filippini et al. 2001). It is noted that Antonow's equation is inadequate as a scientific

approach and is only applicable to calculation of interfacial tensions in complete wetting and non-wetting behaviour (Kwok and Neumann 1999; Yeh et al. 2004).

The molecular weight and the polydispersity index of the homopolymers were characterized by gel permeation chromatography (GPC) performed on a Waters GPC system, equipped with a Waters 1515 isocratic HPLC pump, a Waters 717 plus Autosampler injector, and a Waters 2414 refractive index detector. The eluents were tetrahydrofuran for PPFS and DMF for PHEMA, respectively, at a flow rate of 1.0 ml min⁻¹.

Protein adsorption studies

The resistance of functionalized substrata to protein fouling was evaluated by BSA-FITC protein adsorption. Typically, the uncoated and surface-functionalized glass substrata were rinsed initially with PBS and then placed in the protein solution (1 mg ml⁻¹ PBS solution) at room temperature for 2 h. The substrata were gently washed three times with the PBS solution and the adsorption was imaged with a Leica DMLM fluorescence microscope (Leica Microsystems, Wetzlar, Germany), equipped with an excitation filter of 495 nm and an emission filter of 525 nm. The fluorescence intensity, which was proportional to the surface density of the adsorbed BSA-FITC protein, was quantified using the ImageJ software (National Institutes of Health, Bethesda, MD, USA) (Wu et al. 2012).

Bacterial assays

Two bacterial strains, Gram-negative *Pseudomonas* sp. NCIMB 2021 and Gram-positive *S. aureus*, were used to evaluate the bacterial adhesion characteristics and the bactericidal efficacy of the functionalized substrata. *Pseudomonas* sp. NCIMB 2021 was cultured in nutrient-rich simulated seawater according to the method described previously (Yuan and Pehkonen 2007). *S. aureus* was cultured in tryptic soy broth at 37°C overnight. After incubation, the bacterial suspension was centrifuged to remove the supernatant. *Pseudomonas* sp. NCIMB 2021 and *S. aureus* were washed with simulated seawater and PBS, respectively, and resuspended at a concentration of 10⁷ cells ml⁻¹. Each substratum was immersed in 1 ml of the bacterial suspension at 30°C for 24 h. The adhered bacterial cells were investigated under a scanning electron microscope (SEM, JSM-5600 model, JOEL Co., Tokyo, Japan) after fixing with 3% glutaraldehyde and dehydrating with serial ethanol solutions. Quantification of bacterial adhesion and viability on the

uncoated and modified GS were carried out by the spread plate method (Yang et al. 2011). The results were expressed as viable adherent fractions, defined as the percentages of viable adherent bacterial cells on the functionalized GS surfaces relative to those on the uncoated GS.

Settlement assay of barnacle larvae

The culture of *Amphibalanus amphitrite* (= *Balanus amphitrite* (Evans 2009)) has been reported (Phang et al. 2009; Tan et al. 2010). One ml of filtered seawater containing approximately 50 barnacle cyprids was added to the uncoated glass vials and the glass vials modified by polymer brush coatings. Three replicates were used. After incubation for 24 h and 48 h, the barnacle cyprids which had settled and attached to the vials were counted under a stereoscope.

Fouling test in flowing sand-filtered seawater

The uncoated and polymer-functionalized glass substrata were tested in an open tank with flowing sand-filtered seawater (flow rate $\approx$ 13.5 l min⁻¹). After 1 week, the glass substrata were removed and washed gently three times with 0.2 μ m filtered seawater. The deposits on the surfaces were then investigated by SEM.

Results and discussion

Functional polymer brush coatings via SI-ATRP

In comparison to the biocide-releasing technologies, grafting of non-releasing polymer brushes on material surfaces is a more environmentally-friendly alternative to manage marine and aquatic biofouling. In the present work, functional polymer brush coatings were grafted from the substrata via surface-initiated atom transfer radical polymerization (SI-ATRP) (Figure 1). Glass slides (GS) were firstly activated with piranha solution to generate the hydroxylated surfaces (GS-OH), followed by the immobilization of the silane coupling agent containing the ATRP initiator, 4-(chloromethyl)phenyl trichlorosilane (CTS). Polymer brushes with different wettabilities and charges were subsequently grafted from the CTS coupled-glass substrata via SI-ATRP.

Successful functionalization of the substratum surfaces were first confirmed by static water contact angle measurements (Table 1). The static water contact angle decreased from $57 \pm 3^\circ$ for the uncoated GS surface to $19 \pm 3^\circ$ for the hydroxylated GS-OH surface. The large water contact angle of the uncoated GS surface was due to adsorption of adventitious hydrocarbons (hydrophobic components) under

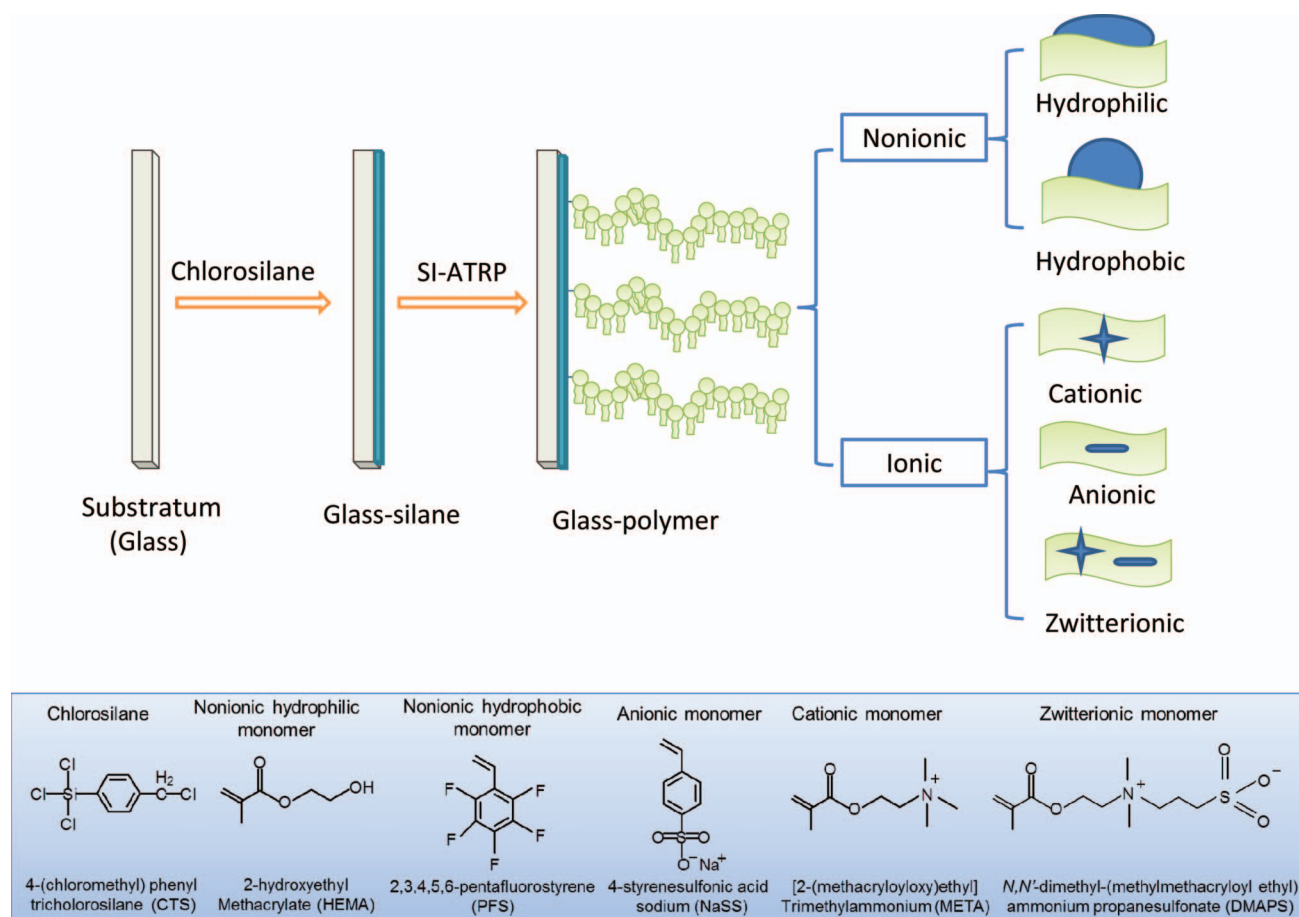


Figure 1. Schematic illustration of the surface-functionalization processes *via* surface-initiated atom transfer radical polymerization of nonionic hydrophilic and hydrophobic, cationic, anionic, and zwitterionic monomers.

ambient condition (Ershad-Langroudi et al. 2009). A pristine glass surface can be obtained, for instance, by cleaning with a gas-air torch, resulting in a very hydrophilic surface with a water contact angle approaching zero degrees. After coupling of CTS, the glass substrata became hydrophobic with a static water contact angle of $85 \pm 2^\circ$. The static water contact angles of the nonionic hydrophilic poly(2-hydroxyethyl methacrylate)-grafted glass (GS-PHEMA) and non-ionic hydrophobic poly(2,3,4,5,6-pentafluorostyrene)-grafted glass (GS-PPFS) surfaces were $41 \pm 2^\circ$ and $95 \pm 3^\circ$, respectively. The cationic, anionic and zwitterionic polymer-modified surfaces were very hydrophilic, having water contact angles of $31 \pm 2^\circ$, $14 \pm 2^\circ$ and $22 \pm 3^\circ$, respectively. The PHEMA homopolymer prepared in one-pot ATRP synthesis had a molecular weight of 61,987 with a polydispersity index (PDI) of 1.18. The PPFS homopolymer had a molecular weight of 27,449 with a PDI of 1.29. These results indicate that well-defined polymer brush coatings were successfully prepared by the surface-initiated

atom transfer radical polymerization. The surface energy for each functionalized surface was calculated from Antonow's equation for water, glycerol and diiodomethane contact angles (Table 1). The hydrophobic GS-PPFS surface showed the lowest surface energy of 32.3 mJ m^{-2} . The hydrophilic GS-PHEMA, GS-PMETA, GS-PNaSS and GS-PDMAPS surfaces had surface energies ranging from 54.7 to 60.7 mJ m^{-2} .

Figure 2 shows the results of XPS analysis of the uncoated, hydroxylated and polymer-grafted GS surfaces. The presence of C 1s signals at a binding energy (BE) of 285 eV (Moulder et al. 1992) in the XPS wide scan spectrum of the uncoated GS surface (inset of Figure 2a), indicated the persistence of residual organic contaminants on the uncoated GS surface prior to activation by piranha solution. The XPS wide scan spectrum of the GS-OH surface (Figure 2a) is only dominated by the oxygen (O 1s) signal at a binding energy (BE) of about 532 eV and silicon (Si 2p and Si 2s) signals at BEs of 99 eV and 151 eV,

Table 1. Surface compositions, contact angles measured with three test liquids (water (θ_w), glycerol (θ_g) and diiodomethane (θ_d)), surface energies, coating thickness and root-mean-square (RMS) roughness of the surface-functionalized glass substrata.

Samples	Surface composition (molar ratio)	Contact angle (mean $\pm$ SD $^\circ$)			Surface energy (mJ m $^{-2}$)		Thickness (nm)	RMS roughness (nm)
		θ_w	θ_g	θ_d	Exp ^b	Ref		
Uncoated GS	—	57 $\pm$ 3	60 $\pm$ 2	42 $\pm$ 3	49.5	48 ^c /53.3 $\pm$ 1.3 ^d	—	0.6
GS-OH	—	19 $\pm$ 3	32 $\pm$ 3	29 $\pm$ 2	59.2	—	—	—
GS-CTS	[C]:[C] = 0.08 (0.14) ^a	85 $\pm$ 2	87 $\pm$ 3	55 $\pm$ 2	37.7	—	5	1.0
GS-PHEMA	[C-H]:[C-O]:[O=C-O] = 2.9:2.2:1 (3:2:1)	41 $\pm$ 2	39 $\pm$ 2	37 $\pm$ 3	54.7	55.9 ^e /44.3 ^f	25	1.7
GS-PPFS	[F]:[C] = 0.46 (0.625)	95 $\pm$ 3	98 $\pm$ 3	65 $\pm$ 3	32.3	32.5 $\pm$ 2.8 ^g	42	2.1
GS-PMETA	[N]:[C] = 0.08 (0.11)	31 $\pm$ 2	37 $\pm$ 2	35 $\pm$ 3	57.1	50.9 ^h /72.4 ^h	30	1.3
GS-PNaSS	[Na]:[S]:[C] = 1:1.03:11.9 (1:1:8)	14 $\pm$ 2	18 $\pm$ 3	27 $\pm$ 2	60.7	72.9 ⁱ	88	0.8
GS-PDMAPS	[S]:[N]:[C] = 1:1.07:14.6 (1:1:11)	22 $\pm$ 3	28 $\pm$ 3	32 $\pm$ 3	59.1	66 $\pm$ 12 ^g /73.1 ^j	90	0.3

Note: ^aExperimental values determined from the XPS analysis. Values in parentheses are the theoretical ratios; ^bExperimental values were the average surface energies determined by the three test liquids; ^cPetrone et al. (2011); ^dChibowski et al. (2002); ^eÇaykara et al. (2003); ^fAricea et al. (2005); ^gWu et al. (2012); ^{h,i,j}Kobayashi et al. (2012).

respectively (Moulder et al. 1992). In comparison to the XPS wide scan spectrum of the GS-OH surface (Figure 2a), additional signals at BEs of about 199, 271 and 284 eV, corresponding to Cl 2p, Cl 2s and C 1s species (Moulder et al. 1992), appear in the XPS wide scan spectrum of the GS-CTS surface (Figure 2b). The appearance of these signals is consistent with the presence of a chlorosilane layer on the glass surface (Zhang et al. 2006). The Si 2p and Si 2s signals, with BEs of about 99 eV and 151 eV, respectively, were not detected in the wide scan spectrum of the GS-PHEMA surface (Figure 2c), indicating that dense PHEMA brushes were grafted on the GS-CTS surface to a thickness exceeding the 8-nm sampling depth of the XPS technique (Tan et al. 1993). Curve fitting of the C 1s core-level spectrum of the GS-PHEMA surface (inset Figure 2c) resulted in three peak components with BEs at about 284.6, 286.2 and 288.5 eV, attributable to C—H, C—O and O—C=O species, respectively (Yang et al. 2011). The area ratio of the three peak components is about 2.9:2.2:0.9, which is in good agreement with the theoretical ratio of 3:2:1 for the PHEMA homopolymer. For the GS-PPFS surface, a strong fluorine signal at a BE of about 688 eV appeared in the wide scan spectrum shown in Figure 2d (Moulder et al. 1992), indicating that PPFS brushes were successfully grafted on the substratum *via* SI-ATRP.

In comparison to the XPS wide scan spectrum of the GS-CTS surface (Figure 2b), a strong nitrogen signal at a BE of about 402 eV was observed in the wide scan spectrum of the cationic GS-PMETA surface (Figure 2e) (Moulder et al. 1992), indicative of the presence of poly(META) brushes on the surface. Only positively charged nitrogen ($-N^+$) species with a BE at 402.1 eV appeared in the N 1s core-level spectrum of the GS-PMETA surface (inset of Figure 2e), consistent with the presence of quaternary ammonium cations ($-N(CH_3)_3^+$) in the PMETA brushes (Moulder et al. 1992). In the wide scan spectrum of anionic GS-PNaSS surface (Figure 2f), the Na 1s and KLL signals (Na 1s: 1072 eV; Na KLL: 493 eV), as well as the S 2p and S 2s signals (S 2p: 164 eV; S 2s: 228 eV) (Moulder et al. 1992; Xu et al. 2005) confirmed the presence of PNaSS brushes on the surface. The [Na]:[S] ratio, as determined from the S 2p and Na 1s core-level spectral area ratio, was about 1:1.03, in good agreement with the theoretical ratio of 1:1 for the chemical structure of NaSS.

For the zwitterionic GS-PDMAPS surface, the S 2p, S 2s and N 1s signals in the wide scan spectrum (Figure 2g) indicated successful graft polymerization of DMAPS from the GS-CTS surface. The positively charged nitrogen ($-N^+$) species at a BE of 402.1 eV in the N 1s core-level spectrum (Figure 2h) is associated

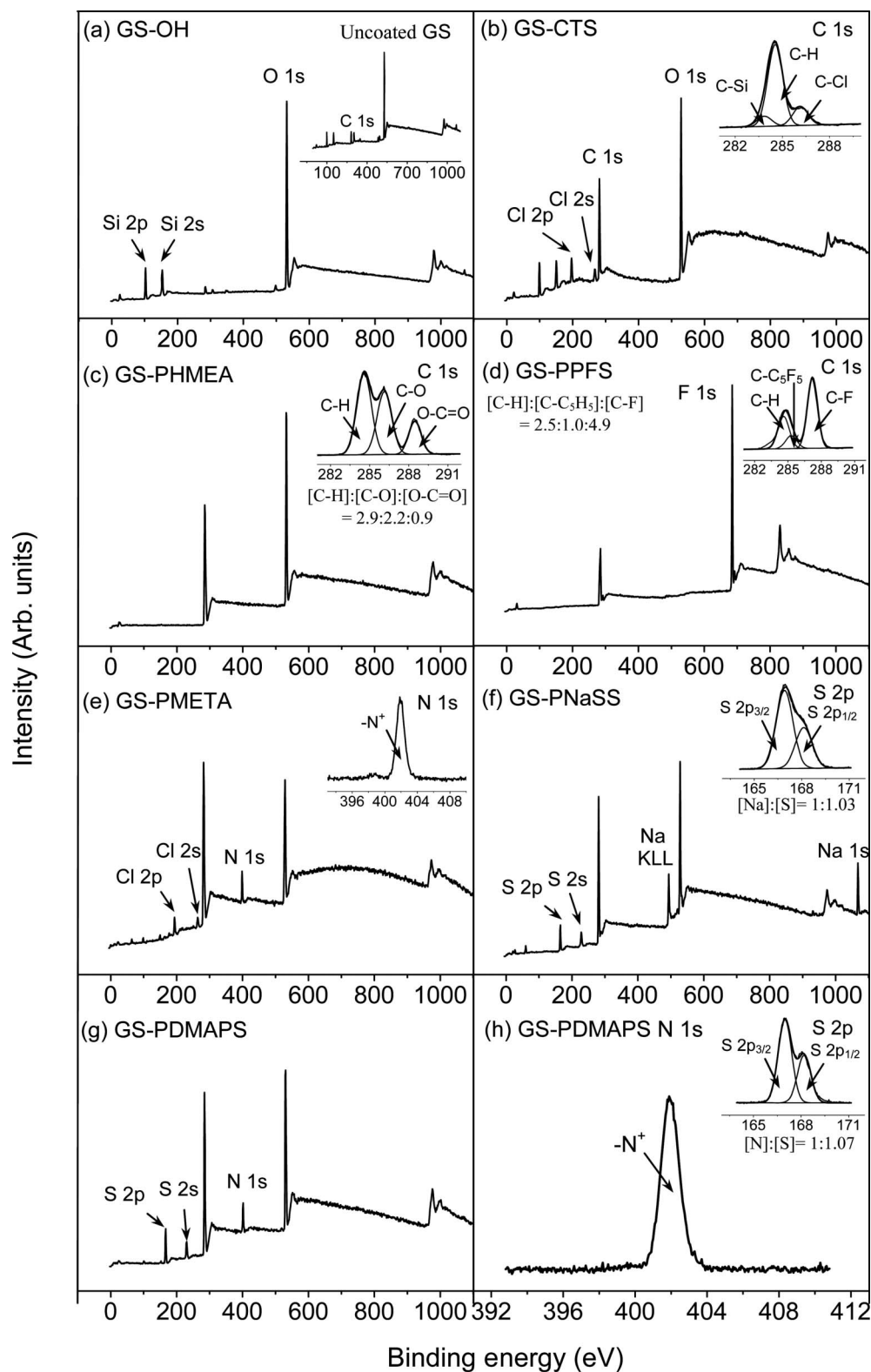


Figure 2. XPS wide scan spectra of the (a) GS-OH (Inset: uncoated GS), (b) GS-CTS, (c) GS-PHEMA, (d) GS-PPFS, (e) GS-PMETA, (f) GS-PNaSS and (g) GS-PDMAPS surfaces. (h) N 1s core-level spectrum of the GS-PDMAPS surface.

with the quaternary ammonium cations ($-\text{N}(\text{CH}_3)_3^+$) of the PDMAAPS brushes. The S 2p_{3/2} and S 2p_{1/2} doublet with BEs at 167.0 and 168.2 eV, respectively (inset of Figure 2h), are associated with the sulfonate groups of PDMAAPS (Moulder et al. 1992; Cai et al. 2011). The [N]:[S] ratio of 1:1.07, derived from the N 1s and S 2p core-level spectra area ratio, was consistent with the theoretical ratio of 1:1 for the PDMAAPS brushes.

The AFM height and phase images in Figure 3 reveal the morphology of the surfaces modified by SI-ATRP in the present work. All graft-modified surfaces had a dense and uniform polymer brush coating (root-mean-square (RMS) roughness: 0.3–2.1 nm over an area of $1 \times 1 \mu\text{m}$). The AFM phase images revealed homogeneity over each graft-modified surface, indicating that the entire surface was uniformly covered by a polymer brush coating. The thickness of polymer layer was between 25 nm and 90 nm, as determined from the surface scratch analysis (upper insets of Figure 3 and Table 1).

Protein adsorption of the functionalized surfaces

Protein adsorption and fouling are closely related to surface wettability and charge (Ostuni et al. 1999; Sethuraman et al. 2004; Xu et al. 2009b). The protein resistance of different polymer brush coatings was assessed by exposure to fluorescein isothiocyanate-conjugated bovine serum albumin (BSA-FITC) in PBS (1 mg ml^{-1}). The protein was chosen due to its well-known strong adsorption characteristics on many types of materials (Chang et al. 2009; Finlay et al. 2010). Figure 4 shows fluorescence microscope images of functionalized substrata with different surface wettability and charges of a different nature after exposure to the protein solution. The average fluorescence intensities of BSA-FITC adsorbed on polymer-grafted surfaces, normalized to that of BSA-FITC on uncoated GS, are shown in Figure 5. After protein adsorption for 2 h, uniform and intense fluorescence was observed on the uncoated GS surface (Figure 4a), indicative of significant BSA adsorption on the glass surface. Extensive adsorption of BSA on hydrophilic surfaces, such as glass (Sweryda-Krawiec et al. 2004; Gudipati et al. 2005) and gold (Kumar et al. 2010; Bhatt et al. 2011) has been reported.

The hydrophilic GS-PHEMA and zwitterionic GS-PDMAAPS surfaces worked well to resist BSA-FITC adsorption. Faint fluorescence was observed on the GS-PHEMA surface (Figure 4b) and a weak fluorescence was visible on the GS-PDMAAPS surface (Figure 4f) after BSA-FITC exposure. Low fouling of oppositely charged protein on a zwitterionic sulfobetaine surface was reported recently (Wu et al. 2012). It has

been reported that a PDMAAPS surface coating 62 nm in thickness gave rise to minimum protein adsorption, and nonspecific protein adsorption increased as the polymer coating thickness increased to 90 nm (Yang et al. 2008). The PDMAAPS brush coating in this study had a thickness of 90 nm, which is not the optimal thickness for resisting protein adsorption, and probably accounts for its weaker resistance to protein adsorption than the coating of PHEMA brushes. A weak fluorescence was discernible on the hydrophobic GS-PPFS surface (Figure 4c), suggesting that the GS-PPFS surface was susceptible to protein fouling arising from hydrophobic interaction between the PPFS coating and BSA-FITC (Chang et al. 2009). Several factors, such as hydrophilicity, surface charges, surface topography, protein concentration and pH, govern the extent of protein adsorption on a surface (Fang and Szleifer 2002; Gudipati et al. 2005; Xu et al. 2009b). In addition, the distance between potential 'contact points' for protein molecules on the surface plays a significant role in defining the extent of protein adsorption onto a surface (Fang and Szleifer 2002; Gudipati et al. 2005). Protein adsorption is minimized or reduced on surfaces that exhibit compositional heterogeneities on length scales similar to those of the protein molecules. Accordingly, the heterogeneity of the GS-PPFS surface (RMS roughness: 2.1 nm) probably accounts for the lower BSA ($14 \times 4 \times 4 \text{ nm}^3$) adsorption on this hydrophobic surface than on the hydrophilic smooth glass surface (RMS roughness: 0.6 nm).

Arising from electrostatic interaction, the weakly negatively-charged BSA (Ba et al. 2010) adsorbed on the cationic surface and was repelled by the anionic surface, as evidenced by the more intense fluorescence from the GS-PMETA surface and weaker fluorescence from the GS-PNaSS surface (Figure 4d and e, respectively). It should be pointed out that this result would be reversed if the anionic and cationic surfaces were to be exposed to a positively-charged protein (Xu et al. 2009b; Ba et al. 2010). Therefore, an effective method to minimize electrostatic interactions and improve resistance to a wide range of proteins should involve the use of a nonionic hydrophilic PHEMA brush coating and a strongly hydrated zwitterionic PDMAAPS brush coating.

Antibacterial assays of the functionalized surfaces

To evaluate the antibacterial efficacy of the polymer brush coatings, the functionalized substrata were exposed to both Gram-negative and Gram-positive bacteria (*Pseudomonas* sp. NCIMB 2021 and *S. aureus*, respectively) for 24 h. Their surface morphologies were subsequently imaged by SEM, as shown in Figure 6.

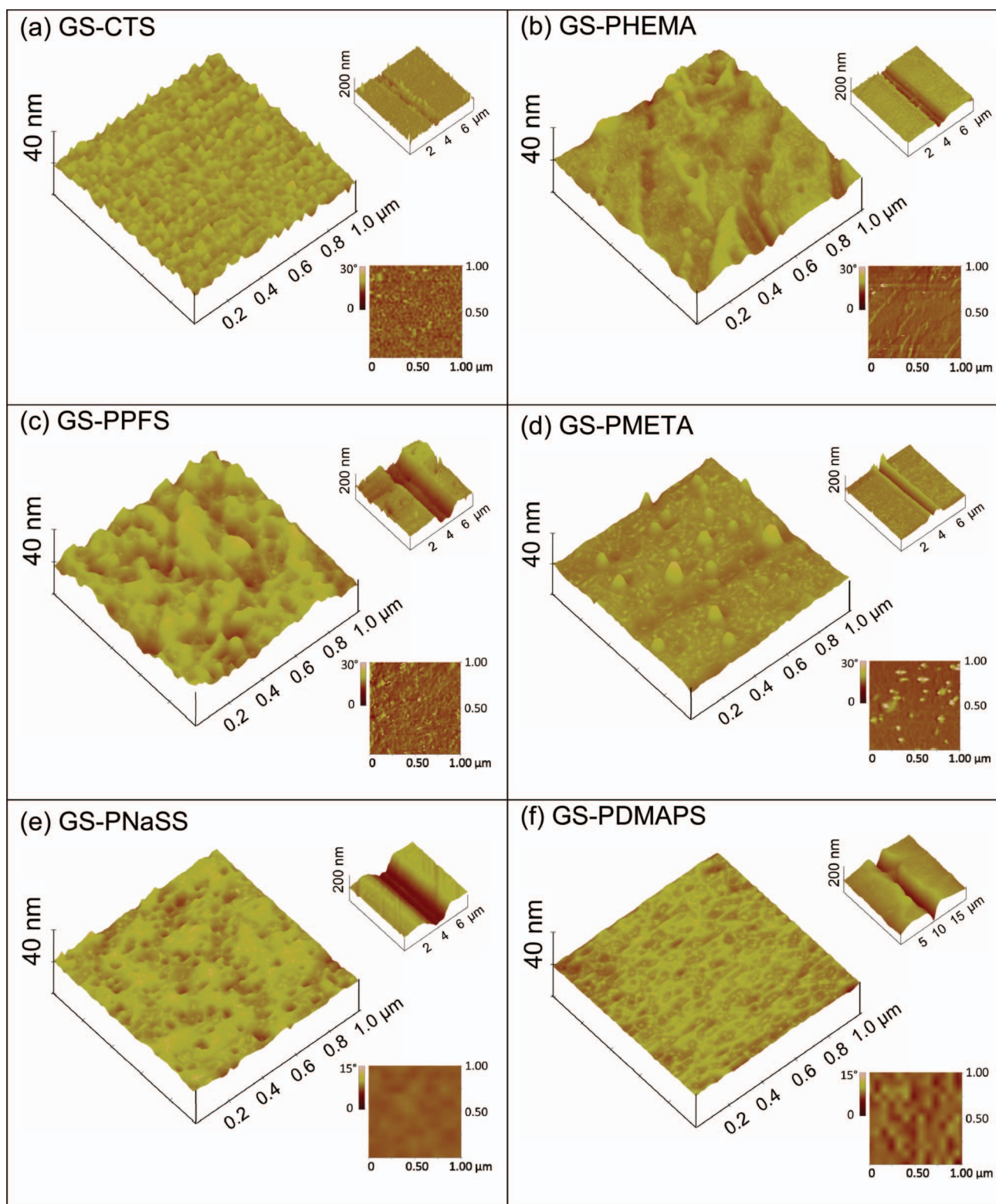


Figure 3. AFM topographical images of the (a) GS-CTS, (b) GS-PHEMA, (c) GS-PPFS, (d) GS-PMETA, (e) GS-PNaSS and (f) GS-PDMAPS surfaces. The AFM phase images are shown in the lower insets (bottom). The AFM images of the scratched surfaces (upper insets) reveal the thickness of the corresponding polymer brush coatings.

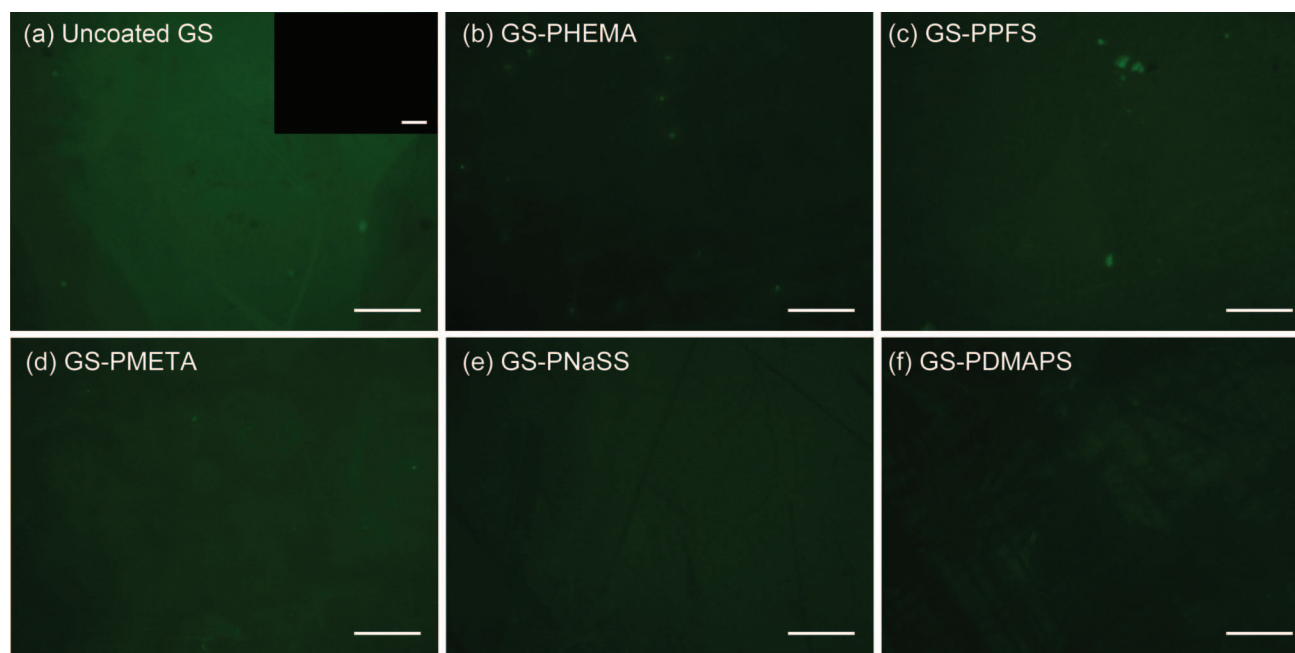


Figure 4. Fluorescence microscope images of the (a) uncoated GS, (b) GS-PHEMA, (c) GS-PPFS, (d) GS-PMETA, (e) GS-PNaSS, (f) GS-PDMAPS surfaces after exposure to PBS solution of BSA-FITC for 2 h. The inset in (a) shows the fluorescence image of the uncoated GS surface before exposure to the BSA-FITC solution. Scale bar = 100 μm .

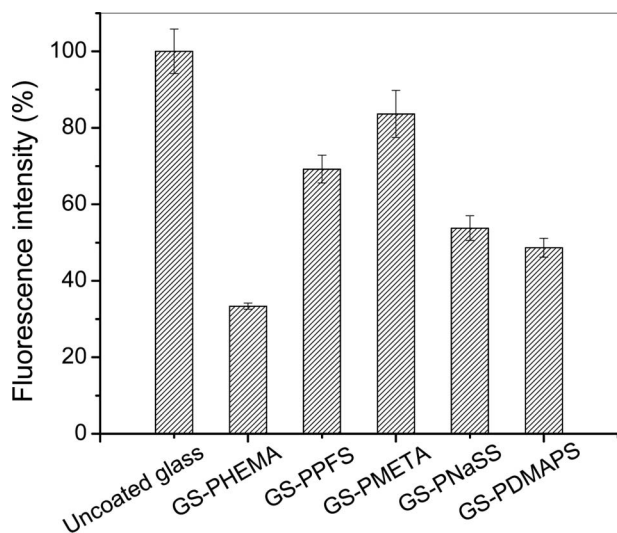


Figure 5. Relative fluorescence intensities of the GS-PHEMA, GS-PPFS, GS-PMETA, GS-PNaSS and GS-PDMAPS surfaces to that of the uncoated GS surface after exposure to the PBS solution of BSA-FITC for 2 h.

After exposure to the bacterial suspension, the bacterial cells adhered readily to the uncoated glass surface (GS), individually and as small clusters (Figure 6a and b), indicating the susceptibility of uncoated GS to bacterial colonization. Similar phenomena were observed on the hydroxylated GS-OH substrata. In

contrast, the hydrophilic GS-PHEMA surface inhibited bacterial adhesion (Figure 6c and d). The decrease in bacterial adhesion on the hydrophobic GS-PPFS surface (Figure 6e and f) indicated its modest resistance to bacterial adhesion.

For the charged surfaces, the cationic GS-PMETA surface was resistant to bacterial adhesion, as evidenced by the small number of bacterial cells adhering to the surface (Figure 6g and h). Bacterial cells killed upon contact with the quaternary ammonium cations might not attach to the surface (Murata et al. 2007; Huang et al. 2008; Wang et al. 2011), could not replicate, and could have degraded in 24 h. Compared to the uncoated GS surface, there was only a small decrease in bacterial adhesion to the anionic GS-PNaSS surface (Figure 6i and j). However, there was a substantial reduction in bacterial adhesion to the zwitterionic GS-PDMAPS surface (Figure 6k and l). The efficacies of charged surfaces in reducing adhesion of *Pseudomonas* sp. NCIMB 2021 and *S. aureus* follow the descending order: zwitterionic > cationic > anionic surface. The lower fouling resistance of the anionic surface compared with the cationic surface for *Pseudomonas* sp. NCIMB 2021 is consistent with that reported for charged hydrogel surfaces (Rasmussen and Ostgaard 2003).

A more quantitative *in vitro* antibacterial assay was carried out using the spread plate method. Figure 7 shows the viable adherent fractions of the Gram-

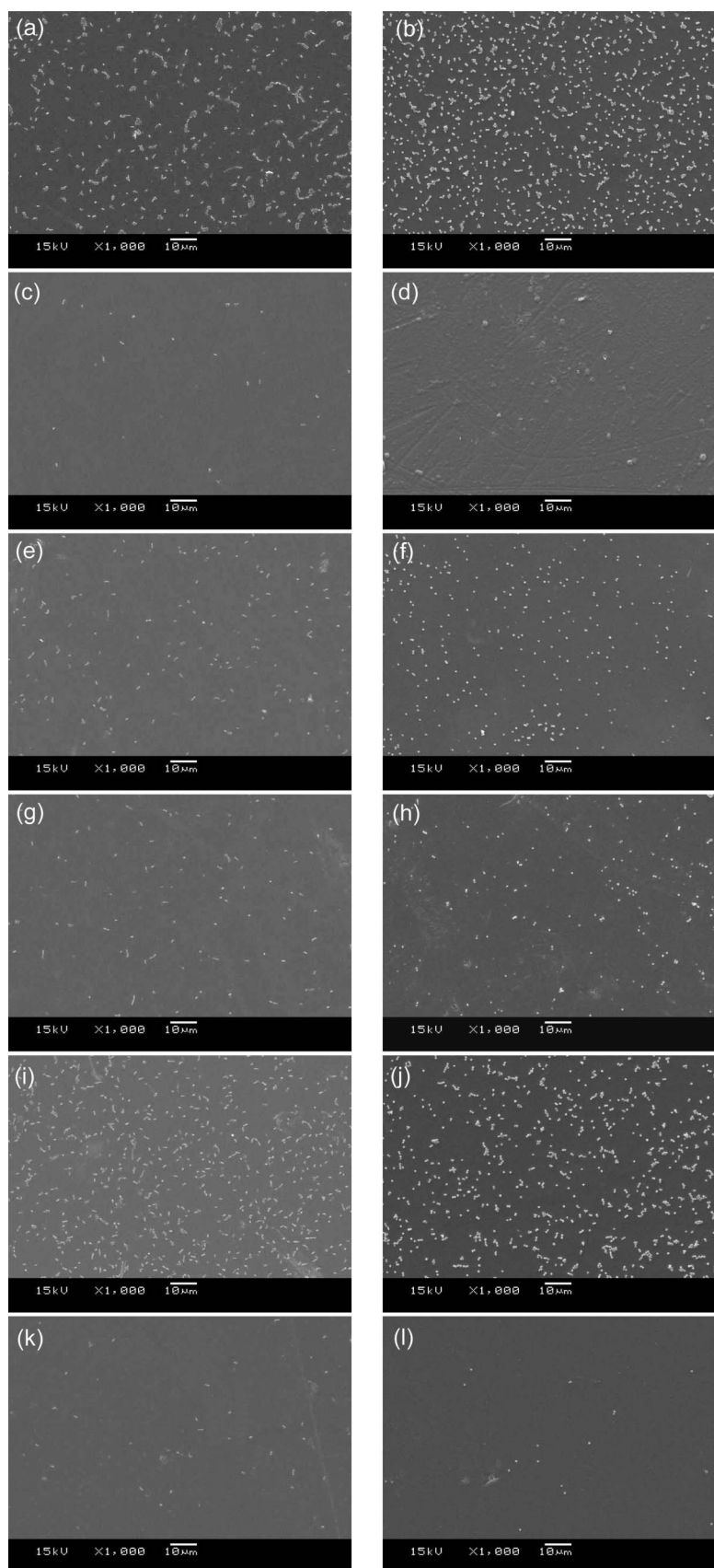


Figure 6. SEM images of the (a,b) uncoated GS, (c,d) GS-PHEMA, (e,f) GS-PPFS, (g,h) GS-PMETA, (i,j) GS-PNaSS and (k,l) GS-PDMAPS surfaces after exposure to *Pseudomonas* sp. NCIMB 2021 (a, c, e, g, i and k) and *S. aureus* (b, d, f, h, j and l) for 24 h.

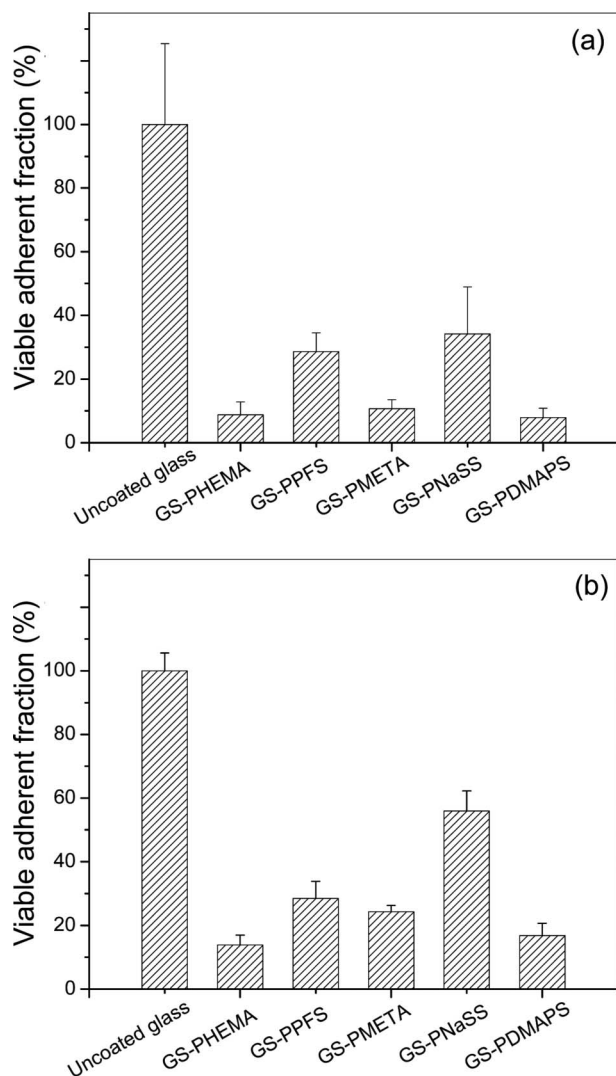


Figure 7. Viable adherent fractions of (a) Gram-negative *Pseudomonas* sp. NCIMB 2021 and (b) Gram-positive *S. aureus* on the uncoated and polymer-grafted glass substrata after exposure for 24 h to the bacterial suspension. The cell number was determined by the spread plate method. Each error bar represents the SD calculated from three replicates.

negative *Pseudomonas* sp. NCIMB 2021 (Figure 7a) and the Gram-positive *S. aureus* (Figure 7b) on the uncoated and functionalized GS surfaces after exposure for 24 h to bacterial suspensions. In comparison to the uncoated GS surface, the polymer-grafted surfaces exhibited more than a 60% reduction in the adherent viability of *Pseudomonas* sp. NCIMB 2021 and a 45% reduction in adhesion of *S. Aureus*. The hydrophilic GS-PHEMA surface exhibited a better anti-bacterial adhesion performance than that of the hydrophobic GS-PPFS surface, as evidenced by the lower adherent viability of both the Gram-negative and the Gram-positive bacteria (Figure 7). The GS-PMETA and GS-PDMAPS surfaces exhibited good

antibacterial ability, with viable adherent fractions of only about 7~10% for *Pseudomonas* sp. NCIMB 2021 and 16~23% for *S. aureus* (Figure 7). The anionic GS-PNaSS surface reduced the adhesion of *Pseudomonas* sp. NCIMB 2021 and *S. aureus* by >60% and 40%, respectively, indicating its comparatively lower anti-bacterial efficiency.

All polymer-grafted substrata with varying degrees of wettability and charges of a different nature reduced the adhesion or viabilities of both Gram-negative *Pseudomonas* sp. NCIMB 2021 and Gram-positive *S. aureus*, as evidenced by the *in vitro* antibacterial tests and SEM images. The variation in antibacterial performance can be explained by hydrophobic/hydrophilic and electrostatic interactions, which have been considered as the two main factors dominating the extent of bacterial adhesion (Webb et al. 1998; Li and Logan 2004). The hydrophilic GS-PHEMA and zwitterionic GS-PDMAPS surfaces showed high efficiency in reducing bacterial adhesion because they can achieve a high degree of hydration in an aqueous environment, which means the polymer 'prefers to be' in contact with water rather than biological matter (Xu et al. 2009a; Jiang and Cao 2010). In comparison to the hydration of nonionic hydrophilic polymer brushes *via* hydrogen-bonding, the electrostatically induced hydration of zwitterionic polymer brushes is more pronounced (Chen et al. 2005; He et al. 2008). Accordingly, the zwitterionic polymers are potentially more effective in the marine environment (Zhang et al. 2009). The hydrophobic GS-PPFS surface had a lower surface energy, resulting in the fouling-release effect and a reduction in bacterial adhesion (Marmur 2006).

Although the charged surfaces were hydrophilic, their anti-bacterial adhesion performances were shown to be more dependent on the nature of the surface charges. The hydrophilicity of the charged surfaces was in the following descending order: the anionic GS-PNaSS surface (water contact angle (CA): 14°) > the zwitterionic GS-PDMAPS surface (CA: 22°) > the cationic GS-PMETA surface (CA: 31°), whereas their anti-bacterial adhesion ability followed the order of the zwitterionic > the cationic > the anionic surface. The hierarchy of antibacterial adhesion can be attributed to the difference in the interaction of bacteria with surface charges. The anionic GS-PNaSS surface resisted bacterial adhesion only by electrostatic repulsion, while the cationic GS-PMETA surface reduced bacterial viability *via* the lethal effect of quaternary ammonium cation groups on the bacteria, with the concomitant reduction in bacterial affinity to the substrata (Murata et al. 2007; Huang et al. 2008; Wang et al. 2011). The zwitterionic GS-PDMAPS surface formed a dense hydration layer from solvation of the negative and positive terminal groups and effectively

inhibited bacterial attachment (Chen et al. 2005; Jiang and Cao 2010). Furthermore, it will be difficult for the biomolecules to find open patches on the zwitterionic surfaces in the presence of uniformly distributed alternating positive and negative charges (Zhang et al. 2009).

Assay of barnacle cyprid settlement on the functionalized surfaces

Barnacles are global macrofoulers and have an extremely wide geographical distribution (Aldred and Clare 2008). In the present work, settlement of cyprids of the barnacle *A. amphitrite* (= *B. amphitrite*) was used to evaluate the anti-macrofouling efficacies of the functionalized surfaces (Figure 8). After 24 h exposure to the medium, 37% of the barnacle cyprids settled on

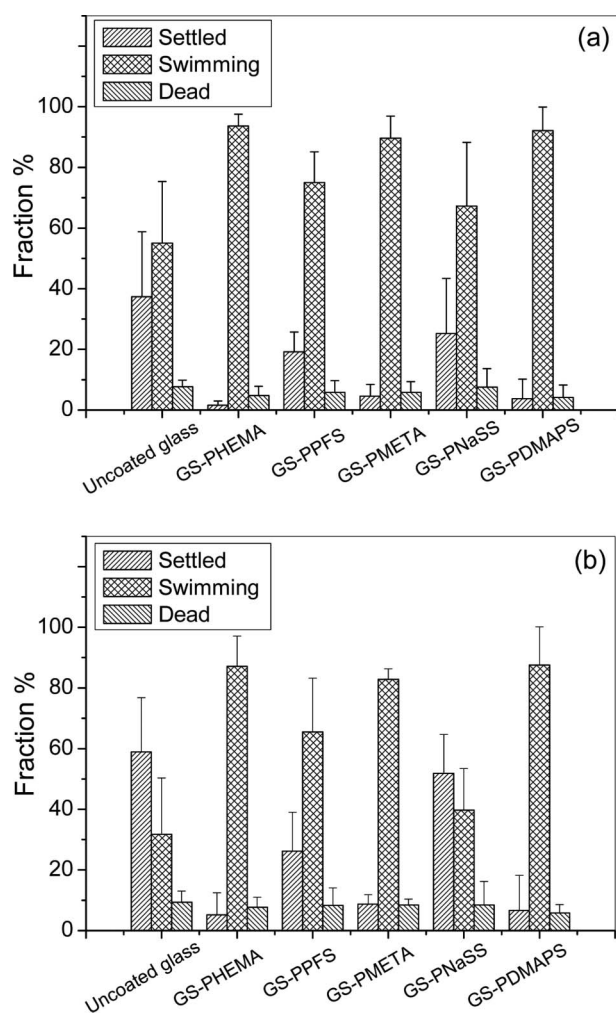


Figure 8. Settled, swimming and dead fractions of barnacle cyprids on the uncoated and polymer-grafted glass substrata after incubation for (a) 24 h and (b) 48 h. Each error bar represents the SD calculated from three replicates.

the uncoated glass surfaces. This number increased to 59% after 48 h. Significant differences in settlement behavior of barnacle cyprids were observed on the different polymer brush coatings. The average fractions of settlement on the hydrophilic GS-PHEMA, cationic GS-PMETA and zwitterionic GS-PDMAPS surfaces were much lower than on the hydrophobic GS-PPFS and anionic GS-PNaSS surfaces after 24 h to 48 h. The extent of settlement was significantly lower on the hydrophilic GS-PHEMA surface than on the uncoated GS surface, with only 5.2% settlement on the GS-PHEMA surface after 48 h. The hydrophobic GS-PPFS surface exhibited a lower efficiency in deterring cyprid attachment than the hydrophilic GS-PHEMA surface, with settlement fractions of 19.2% and 26.2% after 24 h and 48 h, respectively.

As for the charged surfaces, only 8.6% of the cyprids settled on the positively-charged GS-PMETA surface after 48 h, indicating its high resistance to barnacle settlement. The zwitterionic GS-PDMAPS surface reduced barnacle cyprid settlement significantly (6.6% after 48 h). The fractions of settlement on the negatively-charged GS-PNaSS surface were 25.2% and 51.9% after 24 h and 48 h, respectively, only slightly lower than those on the uncoated GS, indicating that the anionic GS-PNaSS surface did not deter barnacle cyprid settlement.

The fractions of cyprid settlement on the polymer-functionalized surfaces are compared to that on the uncoated GS surface in Figure 9. In comparison to the uncoated GS surface, settlement was reduced by up to 92% on the hydrophilic GS-PHEMA surface, and

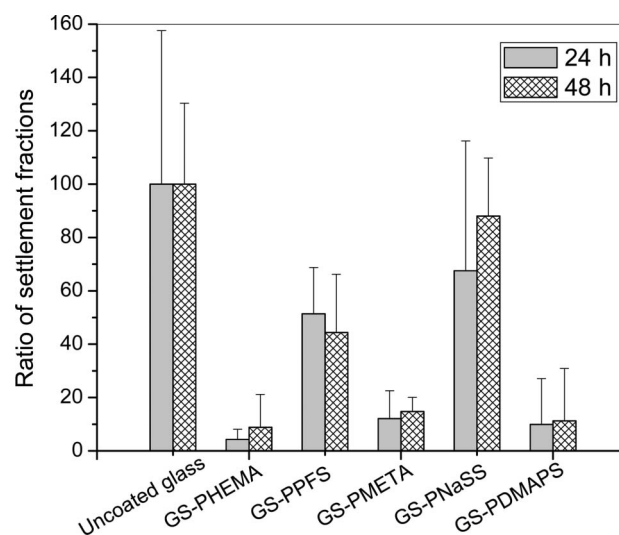


Figure 9. Ratios of settled fractions of barnacle cyprids on the polymer-grafted glass substratum to that on the uncoated GS surface during the 24 and 48 h periods. Each error bar represents the SD calculated from three replicates.

44% on the hydrophobic GS-PPFS surface, after 48 h. Settlement was reduced by >85% on the zwitterionic GS-PDMAPS and the cationic GS-PMETA surfaces compared with that on the uncoated GS surface. The

anionic GS-PNaSS surface exhibited the lowest resistance to the macrofoulers, with a decrease in cyprid attachment of only 12%. A high fraction of barnacle cyprid settlement has also been observed on negatively

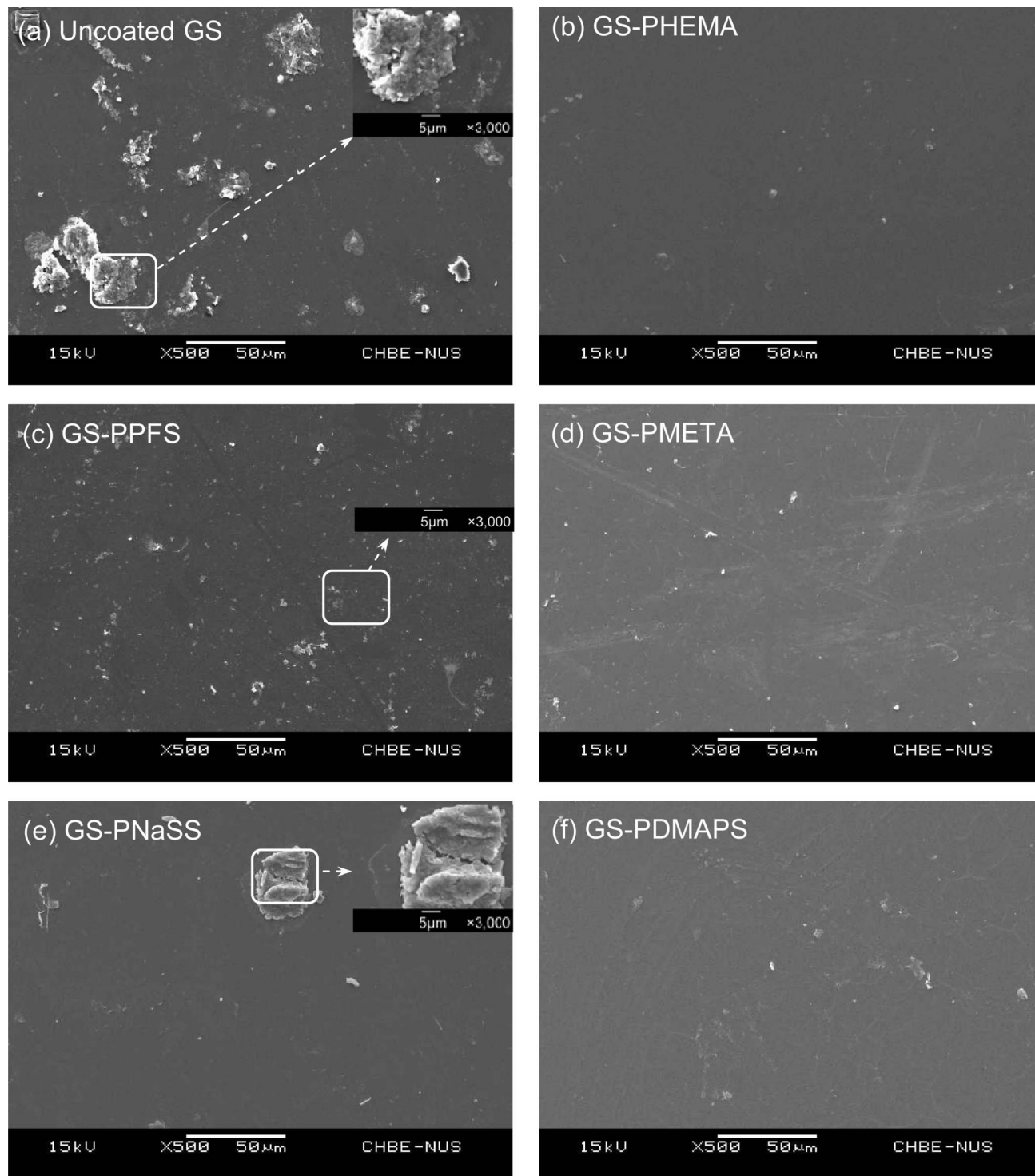


Figure 10. SEM images of the morphologies of the (a) uncoated GS, (b) GS-PHEMA, (c) GS-PPFS, (d) GS-PMETA, (e) GS-PNaSS and (f) GS-PDMAPS surfaces after exposure to flowing seawater (sand-filtered) at 13.5 l min^{-1} for 7 days.

charged surfaces consisting of self-assembled monolayers (SAMs) (Petrone et al. 2011). Four amino acids in barnacle cement, including positively charged lysine and three neutral amino acids, have been reported to be useful for barnacles to couple to foreign material surfaces *via* hydrogen bonding and electrostatic and hydrophobic interaction (Kamino et al. 1996; Kamino 2008). The positively charged lysine in the barnacle cement probably contributed to the attachment of barnacle cyprids to the anionic GS-PNaSS surface *via* electrostatic interaction.

Fouling test by immersion in flow seawater

The uncoated and surface-modified GS were also tested in a flow tank of sand-filtered seawater at a flow rate of 13.5 l min^{-1} for 1 week. Their surface morphologies were subsequently characterized by SEM (Figure 10). The uncoated GS surface was covered extensively by extracellular polysaccharide after immersion for 1 week in the flowing seawater. Clusters of marine bacteria and patchy biofilms are discernible (Figure 10a and inset). In comparison to the uncoated GS surface, marine fouling was significantly reduced on the hydrophilic GS-PHEMA (Figure 10b), the cationic GS-PMETA (Figure 10d) and the zwitterionic GS-PDMAPS (Figure 10f) surfaces. No obvious bacterial clusters or biofilm patches were present on the hydrophobic GS-PPFS surface, albeit the GS-PPFS surface appeared to have been covered by traces of extracellular materials (Figure 10c and inset). A few small bacterial clusters were observed on the anionic GS-PNaSS surface (Figure 10e and inset), indicating the lower fouling resistance of the anionic surface in the absence of heavy hydration and unlike that of the densely hydrated GS-PDMAPS surface. The flow test results also indicate that all the polymer brush coatings were stable and effective in inhibiting biofouling in flowing seawater for 1 week.

Conclusions

Environmentally-benign dense and uniform polymer brush coatings of varying degrees of wettability and with charges of a varying nature were prepared *via* the surface-initiated atom transfer radical polymerization (SI-ATRP) to inhibit marine biofouling. Their AF efficacies were investigated by microfouling and macrofouling assays. The surfaces with grafted polymer brushes showed different degrees of resistance to fluorescein isothiocyanate-conjugated BSA, adhesion of Gram-negative (*Pseudomonas* sp. NCIMB 2021) and Gram-positive (*S. aureus*) bacteria, as well as barnacle (*A. amphitrite* (= *B. amphitrite*)) cyprid settlement. The hydrophilic GS-PHEMA and zwitterionic GS-

PDMAPS surfaces exhibited good AF performance against both microfoulers and macrofoulers. The cationic GS-PMETA surface was effective in deterring barnacle cyprid settlement and also in reducing bacterial adhesion. The hydrophobic GS-PPFS surface showed a moderate ability in inhibiting protein, and bacterial and barnacle cyprid fouling. The anionic GS-PNaSS surface exhibited the least ability in inhibiting fouling by bacterial and barnacle cyprids, among all the polymer-grafted surfaces. The investigation of the AF performances of polymer brush coatings of a different nature is of critical importance to the development of effective AF coatings. Based on the present results, the best approach to combat biofouling is to impart the surface with both (neutral) hydrophilic and zwitterionic properties. The surface-tethered polymer brush coatings prepared by SI-ATRP in the present work, thus, provide an environmentally-friendly and non-releasing alternative to biocide-releasing AF technologies.

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